# INDIAN SPACE ACADEMY

Remote Sensing & GIS Internship

## PROJECT REPORT

Project Code: P2

## Urban Heat Island Mapping using Land Surface Temperature, Built-up Density, and GIS Analysis

Submitted By

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# 1. Objective

The main objective of this project is to study the Urban Heat Island (UHI) effect in Chennai using satellite remote sensing data. The project aims to:

- Calculate Land Surface Temperature (LST) from Landsat-8 imagery and identify areas with higher surface temperatures.
- Analyze the relationship between built up areas and temperature using the Normalized Difference Built up Index (NDBI).
- Identify urban thermal hotspots and understand how rapid urban development influences surface temperature patterns.

# 2. Study Area

The study area selected for this project is Chennai, located in the state of Tamil Nadu, India. Chennai is one of the fastest growing metropolitan cities in the country, with a high population density, extensive built up development, and a tropical climate characteristics that make it an ideal region for studying the Urban Heat Island effect.

A rectangular Area of Interest (AOI) covering Chennai and its surrounding urban regions was created using the Geometry tool in Google Earth Engine. The selected area includes residential, commercial, industrial, coastal, and vegetated regions, enabling a comprehensive analysis of surface temperature variation and built up density. The AOI was used to extract Landsat-8 satellite imagery for performing LST, NDVI, and NDBI analyses.

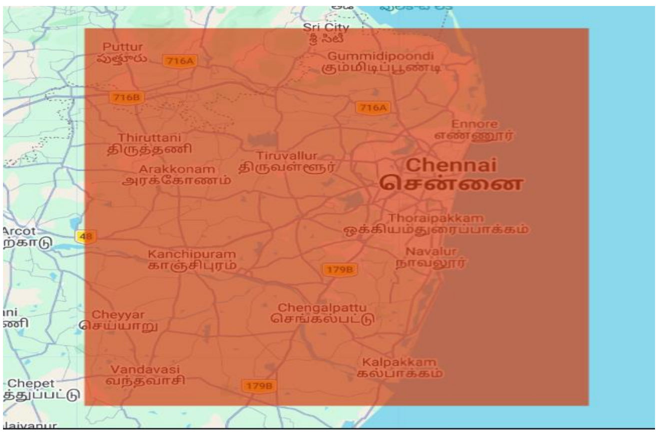


*Figure 1. Area of Interest (AOI) showing the selected study area in Chennai, Tamil Nadu, India.*

# 3. Data Used

The analysis was carried out using Landsat-8 satellite imagery processed in Google Earth Engine (GEE). Images were filtered to obtain cloud free observations over the study area.

## Satellite Data

| Parameter | Details |
|---|---|
| **Satellite** | Landsat-8 OLI/TIRS Collection 2 Level 2 |
| **Study Area** | Chennai, Tamil Nadu, India |
| **Platform** | Google Earth Engine (GEE) |
| **Date Range** | 01 January 2023 – 31 December 2023 |
| **Cloud Cover** | Less than 10% |

## Bands Used

| Band | Description | Purpose |
|---|---|---|
| Band 4 (Red) | Red wavelength | NDVI calculation |
| Band 5 (NIR) | Near Infrared | NDVI calculation |
| Band 10 (Thermal Infrared) | Thermal band | Land Surface Temperature (LST) calculation |

## Derived Products

- Normalized Difference Vegetation Index (NDVI)
- Normalized Difference Built up Index (NDBI)
- Land Surface Temperature (LST)
- Urban Heat Island (UHI) Zones

## Data Sources

USGS EarthExplorer – Landsat-8 Data

Google Earth Engine

Google Earth Engine Code Editor

GEE Data Catalog – Landsat 8 Collection 2 Level-2

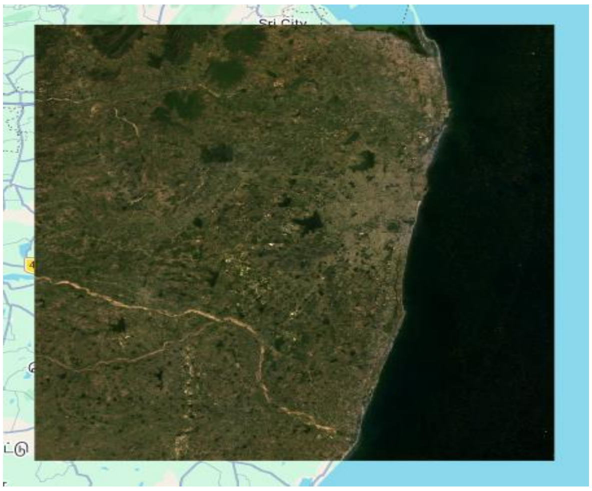

*Figure 2. Landsat-8 RGB image of the selected study area.*

# 4. Methodology

## 4.1 QGIS Based Workflow

The complete analysis was carried out using Google Earth Engine (GEE); QGIS was not used for image processing or analysis in this project. However, the exported raster outputs from GEE can be imported into QGIS for visualization, cartographic map preparation, and further spatial analysis if required.

## 4.2 Google Earth Engine (GEE) Workflow

The complete Urban Heat Island (UHI) analysis was performed using Google Earth Engine. Landsat-8 satellite imagery was processed to calculate NDVI, NDBI, Land Surface Temperature (LST), and Urban Heat Island zones. The workflow is described step by step below.

**Step 1: Selection of Area of Interest (AOI)**

A rectangular Area of Interest (AOI) covering Chennai and its surrounding regions was created using the Geometry tool in Google Earth Engine. This AOI was used to clip the satellite imagery for further analysis.

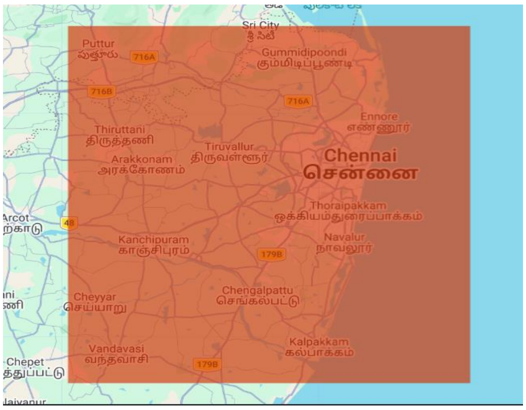


*Figure 3. Area of Interest (AOI) created in Google Earth Engine.*

### Step 2: Loading Landsat-8 Satellite Data

Cloud free Landsat-8 OLI/TIRS Collection 2 Level 2 satellite images for the year 2023 were loaded into Google Earth Engine. The image collection was filtered by AOI, date range, and a cloud cover threshold of less than 10%. A median composite image was then created and clipped to the AOI for further analysis.

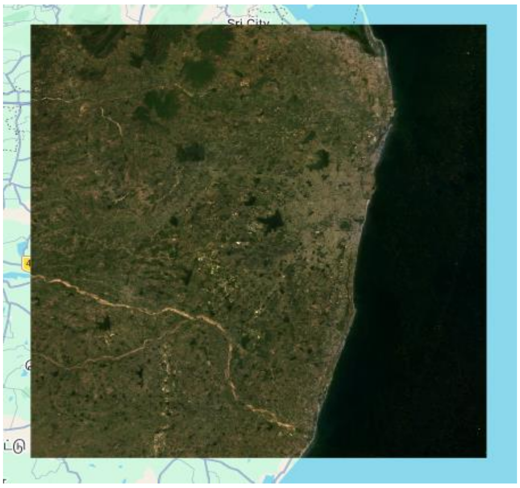

*Figure 4. Landsat-8 RGB composite of the study area used for Urban Heat Island analysis.*

### Step 3: Normalized Difference Vegetation Index (NDVI) Calculation

NDVI was calculated to identify vegetation cover within the study area. It uses the Near Infrared (NIR) and Red bands of Landsat-8 imagery to measure vegetation health and density. Higher NDVI values indicate healthy vegetation; lower values represent barren land, built up areas, or water bodies.

$$NDVI = (NIR - Red) / (NIR + Red)$$

For Landsat-8, Band 5 (NIR) and Band 4 (Red) were used. The resulting NDVI map was displayed using a color palette ranging from low to high vegetation density.

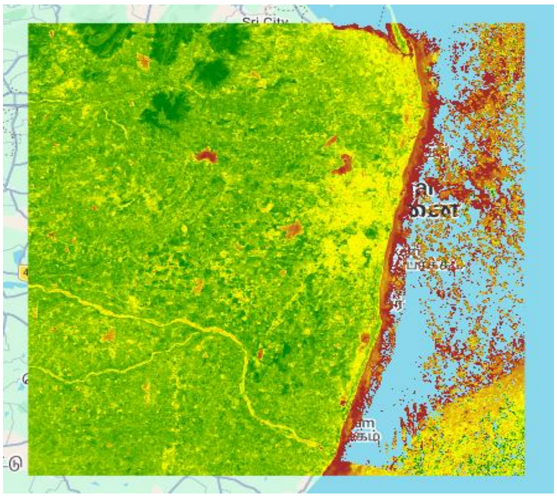

*Figure 5. NDVI map of the study area showing the distribution of vegetation.*

**Step 4: Normalized Difference Built up Index (NDBI) Calculation**

NDBI was calculated to identify built up areas within the study region. It uses the Short Wave Infrared (SWIR) and Near Infrared (NIR) bands of Landsat-8 to distinguish urban surfaces from vegetation and other land cover types.

$$NDBI = (SWIR - NIR) / (SWIR + NIR)$$

Band 6 (SWIR) and Band 5 (NIR) were used. Higher NDBI values indicate urban or built up regions; lower values represent vegetation and water bodies.

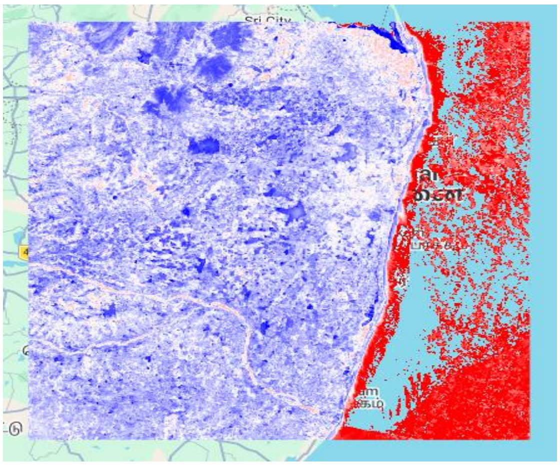

*Figure 6. NDBI map showing the distribution of built up areas in the study area.*

### Step 5: Land Surface Temperature (LST) Calculation

LST was calculated using the thermal infrared band (Band 10) of Landsat-8 imagery in Google Earth Engine. Temperature values were converted from Kelvin to degrees Celsius for easier interpretation. The LST map was visualized using a color gradient from blue (cooler) to red (warmer), providing a basis for UHI analysis.

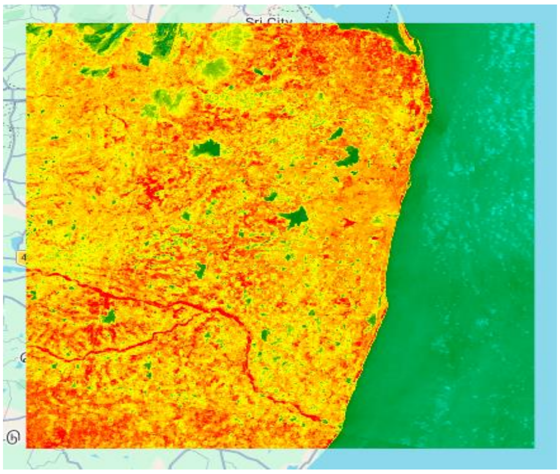

*Figure 7. Land Surface Temperature (LST) map of the study area showing the spatial distribution of surface temperature.*

### Step 6: Urban Heat Island (UHI) Identification

UHI zones were identified using the calculated LST map. A temperature threshold of 40°C was applied pixels with LST values greater than 40°C were classified as Urban Heat Island zones. The resulting map highlights areas with intense heat associated with dense built up regions, limited vegetation, and extensive impervious surfaces.

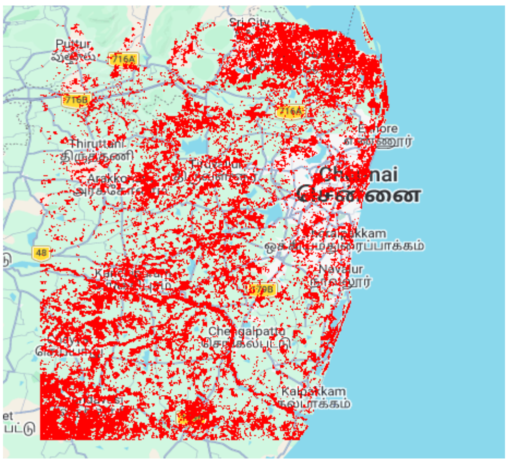


*Figure 8. Urban Heat Island (UHI) zones identified using a temperature threshold of 40°C.*

### Step 7: Built up Density and Temperature Relationship

The relationship between built up density and land surface temperature was examined by comparing the NDBI and LST layers. Regions with higher built up density generally exhibited higher land surface temperatures, indicating the presence of the UHI effect. This analysis demonstrates that increasing urbanization and impervious surfaces contribute to elevated surface temperatures.

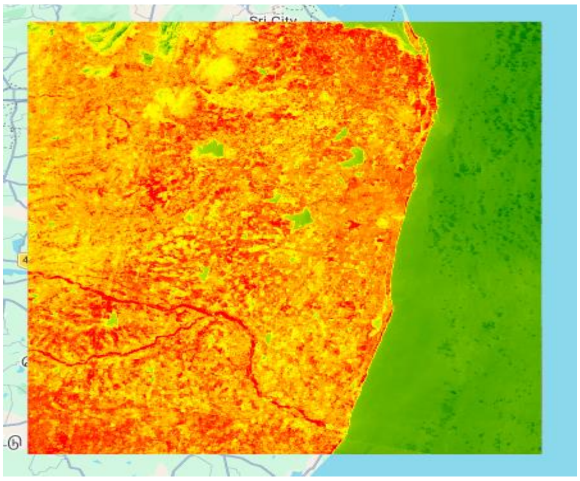

*Figure 9. Built up Density (NDBI) and Land Surface Temperature (LST) relationship within the Chennai study area.*

### Step 8: Spatial Analysis

A spatial analysis was carried out using the generated LST, UHI, and NDBI maps. The analysis showed that higher temperature zones were mainly concentrated in densely built up urban regions, while comparatively lower temperatures were observed in areas with vegetation and water bodies. Comparing these thematic layers helped identify urban thermal hotspots and understand the influence of built up areas on surface temperature variation.

### Step 9: Export of Results

The final LST map and UHI zones were exported from Google Earth Engine as GeoTIFF (.tif) files for documentation and future spatial analysis. The exported raster datasets preserve the original spatial resolution and geographic reference, enabling use in GIS software for visualization and map preparation.

The following output files were exported:

- Chennai_LST.tif – Land Surface Temperature Map
- UHI_Zones_Chennai.tif – Urban Heat Island Zones

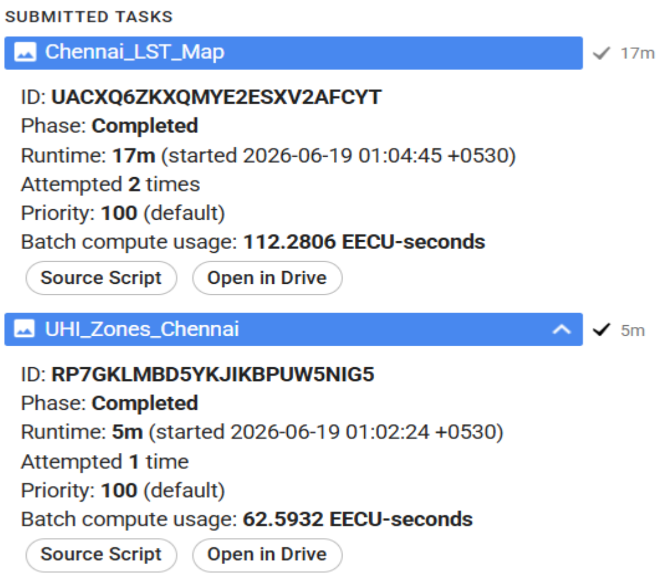


*Figure 10. Export of LST and UHI results from Google Earth Engine.*

### Step 10: Area Calculation

LST statistics and the total UHI area were calculated using Google Earth Engine. The minimum, mean, and maximum LST values provide an overview of the temperature distribution. Additionally, the total area classified as UHI (LST > 40°C) was computed using pixel area statistics and expressed in square kilometres, supporting quantitative assessment of thermal hotspots.

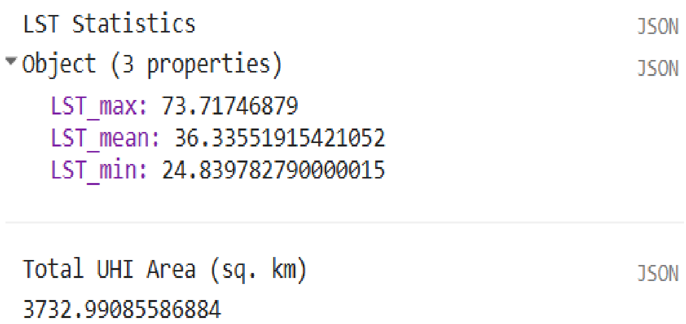


*Figure 11. Console output showing Land Surface Temperature (LST) statistics and Urban Heat Island (UHI) area calculation.*

## 5. Results

The Urban Heat Island analysis successfully generated thematic maps and statistical outputs for the Chennai study area. The results include vegetation distribution (NDVI), built up density (NDBI), land surface temperature (LST), Urban Heat Island (UHI) zones, and area statistics.

## 5.1 Normalized Difference Vegetation Index (NDVI)

The NDVI map represents the spatial distribution of vegetation across the Chennai study area. Higher NDVI values indicate healthy vegetation, while lower values correspond to built up areas, barren land, and other non vegetated surfaces.

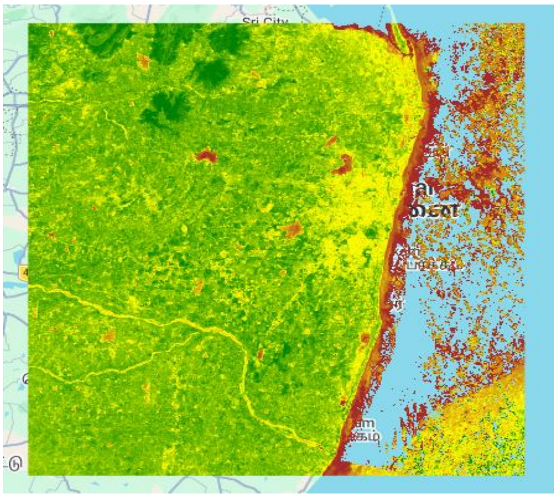

*Figure 12. NDVI map of the Chennai study area.*

### Key Observations

- Higher NDVI values are observed in vegetated and green areas of the study region.
- Lower NDVI values are concentrated in densely built up and barren land areas.
- Vegetated regions generally exhibit lower land surface temperatures due to the cooling effect of vegetation.
- The NDVI map effectively distinguishes vegetation from urban land cover and supports further UHI analysis.

## 5.2 Built up Density (NDBI)

The NDBI map was used to identify built up and urbanized areas within the Chennai study area. Higher NDBI values indicate densely developed regions; lower values represent vegetation, water bodies, and other non built up surfaces.

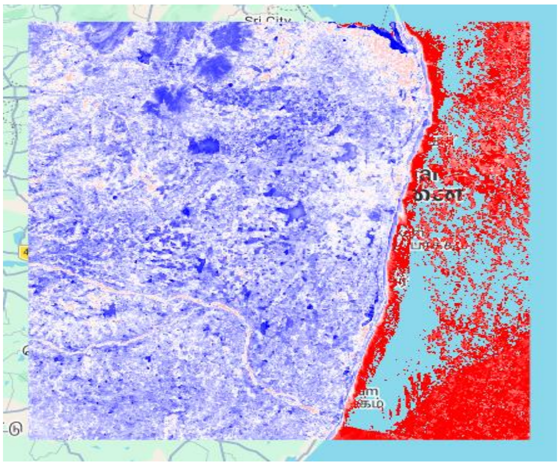

*Figure 13. Built up Density (NDBI) map of the Chennai study area.*

### Key Observations

- Higher NDBI values are concentrated in densely built up urban regions.
- Lower NDBI values correspond to vegetation, open land, and water bodies.
- The NDBI map clearly distinguishes urban development from non urban areas.
- Built up regions identified by NDBI generally correspond to areas with higher land surface temperatures, supporting the UHI analysis.

## 5.3 Land Surface Temperature (LST)

The LST map illustrates the spatial variation in surface temperature across the Chennai study area. The temperature distribution highlights warmer and cooler regions, providing a basis for identifying UHI zones.

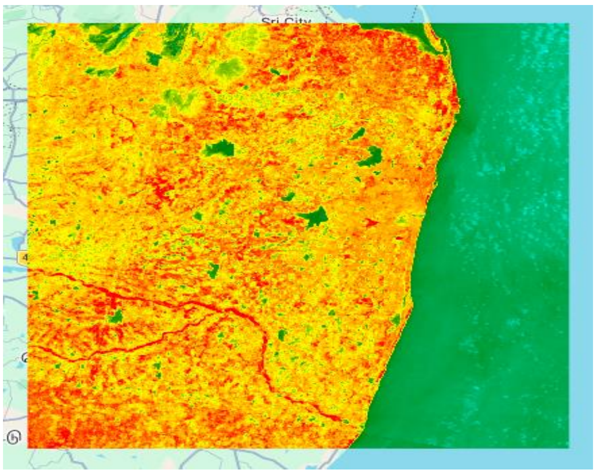

*Figure 14. Land Surface Temperature (LST) map of the Chennai study area.*

### Key Observations

- Surface temperatures range from 24.84°C to 73.72°C, with a mean temperature of 36.34°C.
- Higher temperatures are mainly observed in densely built up and urbanized regions.
- Lower temperatures are associated with vegetated areas and other natural land cover.
- The LST map serves as the primary input for Urban Heat Island analysis.

## 5.4 Urban Heat Island (UHI) Zones

The UHI map identifies areas where Land Surface Temperature exceeds the selected threshold of 40°C. These regions represent thermal hotspots associated with dense urban development and reduced vegetation cover.

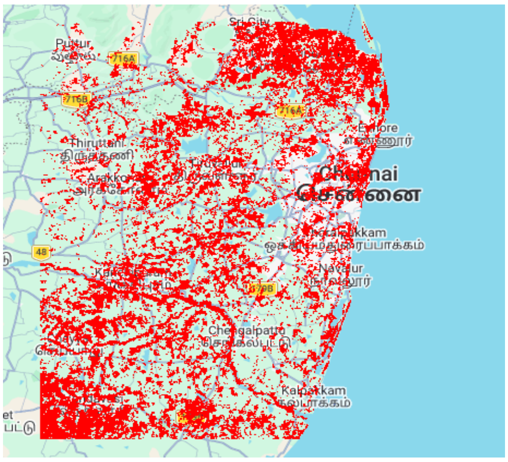


*Figure 15. Urban Heat Island (UHI) zones identified using a temperature threshold of 40°C.*

### Key Observations

- Areas with LST greater than 40°C were classified as Urban Heat Island zones.
- Thermal hotspots are primarily concentrated in densely built up urban regions.
- The total Urban Heat Island area was estimated to be 3,732.99 sq. km.
- The UHI map clearly highlights the spatial distribution of high temperature zones, supporting urban planning and environmental management.

## 5.5 Land Surface Temperature Statistics and Urban Heat Island Area

The statistical analysis of the LST map provides a quantitative summary of temperature distribution across the Chennai study area.

*Table 1. Land Surface Temperature Statistics and Urban Heat Island Area*

| Parameter | Value |
|---|---|
| **Minimum LST** | 24.84°C |
| **Mean LST** | 36.34°C |
| **Maximum LST** | 73.72°C |
| **Total UHI Area** | 3,732.99 sq. km |


```
LST Statistics                                JSON
▾Object (3 properties)                         JSON
  LST_max: 73.71746879
  LST_mean: 36.33551915421052
  LST_min: 24.839782790000015

Total UHI Area (sq. km)                       JSON
3732.99085586884
```


*Figure 16. Google Earth Engine console output showing LST statistics and total UHI area.*

### Key Observations

- The minimum recorded LST was 24.84°C, indicating relatively cooler regions within the study area.
- The mean LST of 36.34°C suggests that a significant portion of the area experiences moderately high temperatures.
- The maximum LST reached 73.72°C, highlighting the presence of intense thermal hotspots.
- The total UHI area was estimated at 3,732.99 sq. km, indicating extensive heat affected regions within the selected study area.
- These statistical results validate the spatial analysis and support the identification of urban thermal hotspots.

## 5.6 Summary of Findings

The analysis of the Chennai study area led to the following key findings:

- Vegetated regions exhibited higher NDVI values and comparatively lower land surface temperatures.
- Built up areas identified using the NDBI map generally showed higher surface temperatures than surrounding regions.

- The LST map revealed significant spatial variation, with temperatures ranging from 24.84°C to 73.72°C.
- Urban Heat Island zones were successfully identified using a threshold temperature of 40°C, covering a total area of 3,732.99 sq. km.
- The results indicate a positive relationship between urban development and increasing land surface temperature, demonstrating the influence of built up areas on the Urban Heat Island effect.

# 6. Conclusion

This project successfully analyzed the Urban Heat Island (UHI) effect in the Chennai study area using Google Earth Engine and Landsat-8 satellite imagery. The generated NDVI, NDBI, Land Surface Temperature, and UHI maps effectively identified vegetation cover, built up regions, temperature distribution, and urban thermal hotspots.

The results showed that densely built up areas generally experienced higher land surface temperatures, while vegetated regions exhibited comparatively lower temperatures. A total Urban Heat Island area of 3,732.99 sq. km was identified using a threshold temperature of 40°C, demonstrating the clear relationship between urban development and increasing surface temperature.

Although the analysis was based on a single year of satellite imagery and a fixed temperature threshold, the methodology can be further improved by incorporating multi season datasets, higher resolution satellite imagery, and additional environmental parameters. Overall, the project successfully achieved its objective of identifying high temperature zones and analyzing their relationship with built up density, providing valuable insights for urban planning and environmental management.

# 7. References

[1] Google Earth Engine. https://earthengine.google.com/

[2] Google Earth Engine Code Editor. https://code.earthengine.google.com/

[3] United States Geological Survey (USGS) EarthExplorer. https://earthexplorer.usgs.gov/

[4] Google Earth Engine Data Catalog – Landsat 8 Collection 2 Level 2. https://developers.google.com/earth-engine/datasets/catalog/LANDSAT_LC08_C02_T1_L2